



Network+ Essentials: Concepts, Math, and Real-World Applications

Introduction

Purpose

Understand Local Area Network(LAN) components, switching, duplexing, troubleshooting, and performance metrics.



Approach

Break down each networking concept

Show the **formula**

Explain **why it matters**

Give examples

LAN Components

Concepts

- User devices, switches, routers, NICs, cables, Ethernet/IP protocols.
- IP addressing & subnetting basics.



Key Formulas

Bandwidth Utilization:

Hosts in /24:

$$2^{(32-24)} - 2 = 254$$

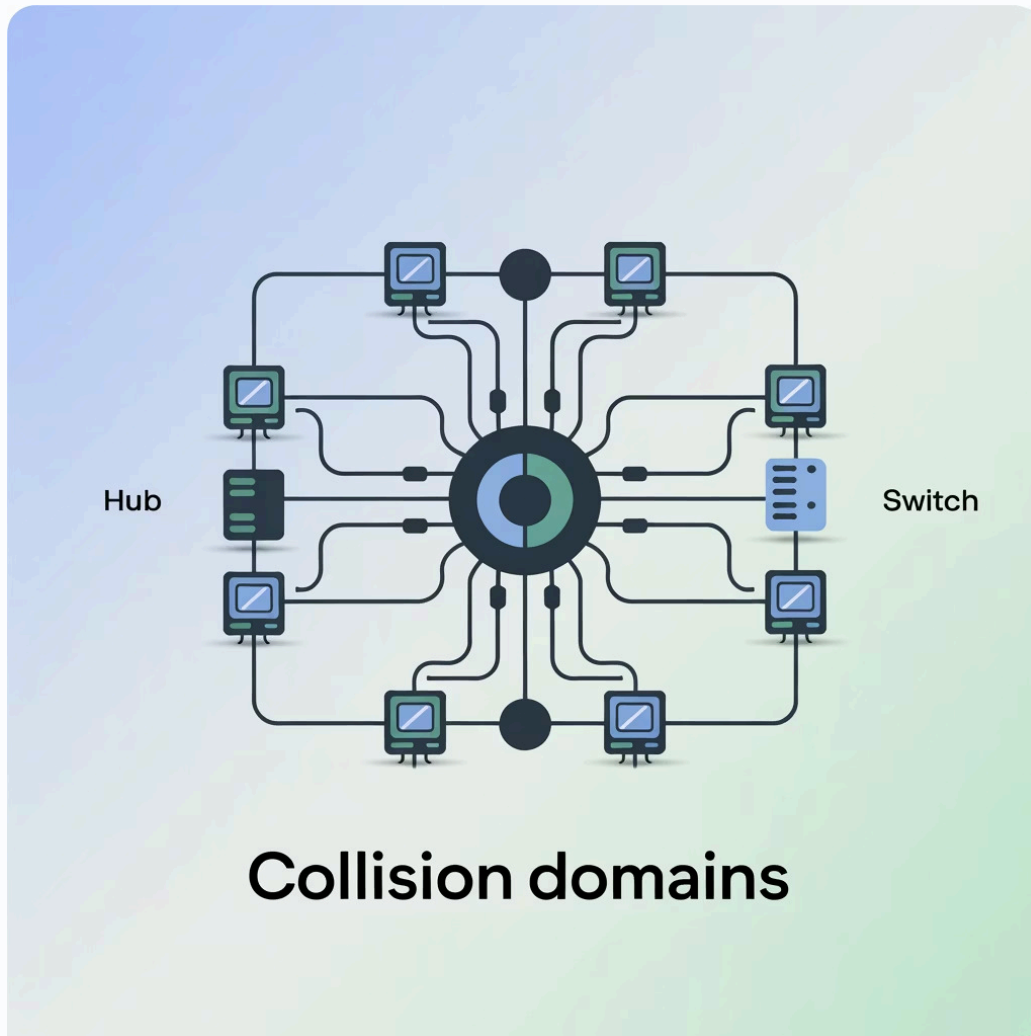
Example

800 Mbps / 1 Gbps → **80% utilization** → upgrade needed.

Collision Domains

Concepts

- Hubs: one shared domain.
- Switch ports: separate domains.



Formula

Collision Probability:

$$P_c = 1 - \left(1 - \frac{1}{n}\right)^{n-1}$$

Example

10 devices on a hub → ~61% collision chance.

Layer 2 Switch Functions

Concepts

- MAC-based forwarding.
- Full-duplex capability.



Formula

Switch Capacity:

$$\text{Capacity} = \text{Ports} \times \text{Speed} \times 2$$

Example

48 ports $\times$ 1 Gbps full-duplex = **96 Gbps**.

Router vs Switch Ports

Concepts

- Routers: fewer, costlier ports.
- Switches: more, cheaper ports.

Formula

Cost/Port:

$$\frac{\text{Cost}}{\text{Ports}}$$

Example

Switch: \$25/port vs Router: \$100/port.



Switch LED Indicators

Concepts

- Link, speed, duplex, errors.

Formula

Utilization:

Example

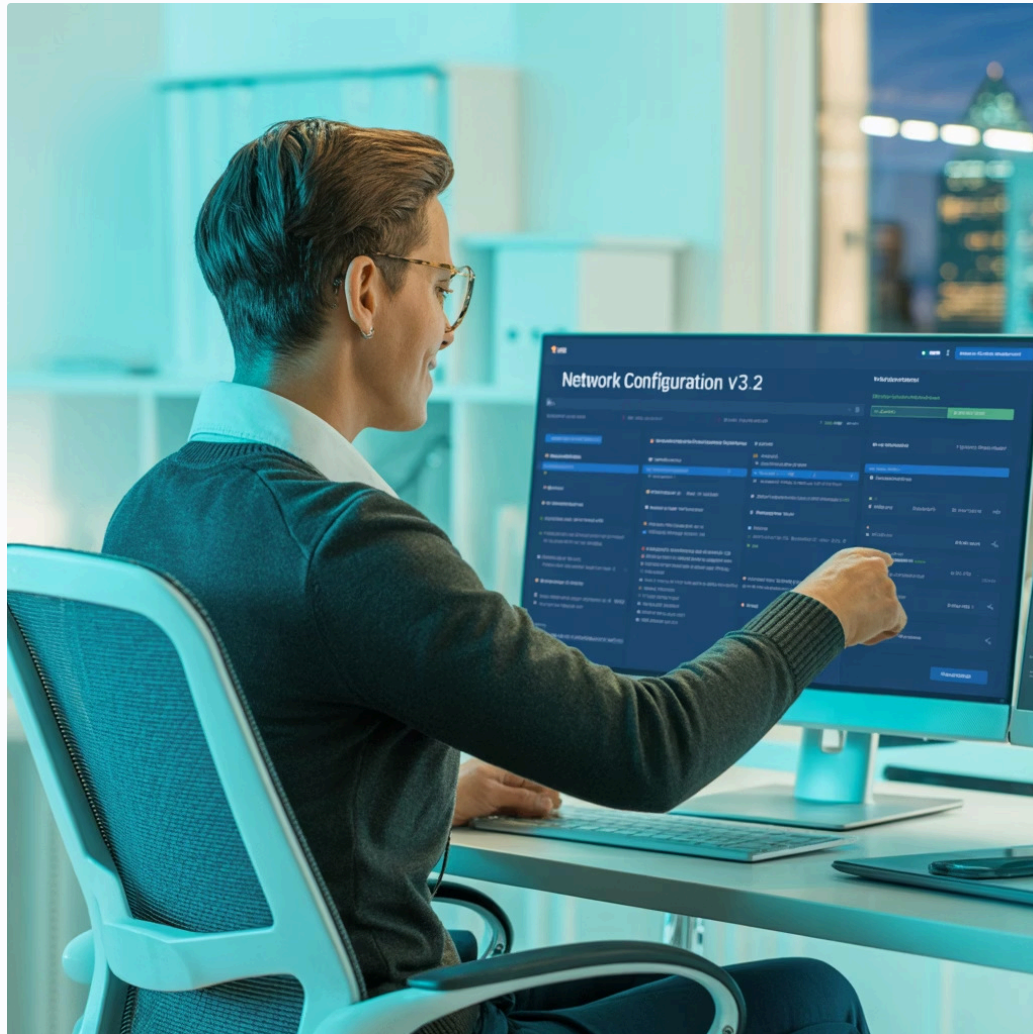
500 Mbps / 1 Gbps = **50% utilization.**



Management IP

Concepts

- VLAN interface for remote management.



Formula

Hosts in /24:

$$2^8 - 2 = 254$$

Example

IP 172.20.137.5 → Gateway test via ping.

Duplex & Speed

Concepts

- Half-duplex = send or receive.
- Full-duplex = both simultaneously.

Formula

Full-Duplex Throughput:

$$2 \times \text{Link Speed}$$

Example

1 Gbps full-duplex = **2 Gbps total**.



MAC Address Learning

Concepts

- Switch CAM table → stores learned MACs.



Formula

Hit Rate:

$$\frac{\text{Known Lookups}}{\text{Total}} \times 100$$

Example

800/1000 lookups = **80% hit rate**.

Troubleshooting Tools

Concepts

- Ping (RTT & packet loss).
- Telnet (L7 testing).
- Interface status (show interface).



Formulas

Avg RTT:

$$\frac{\sum RTT}{n}$$

Error Rate:

$$\frac{\text{Errors}}{\text{Total Frames}} \times 100$$

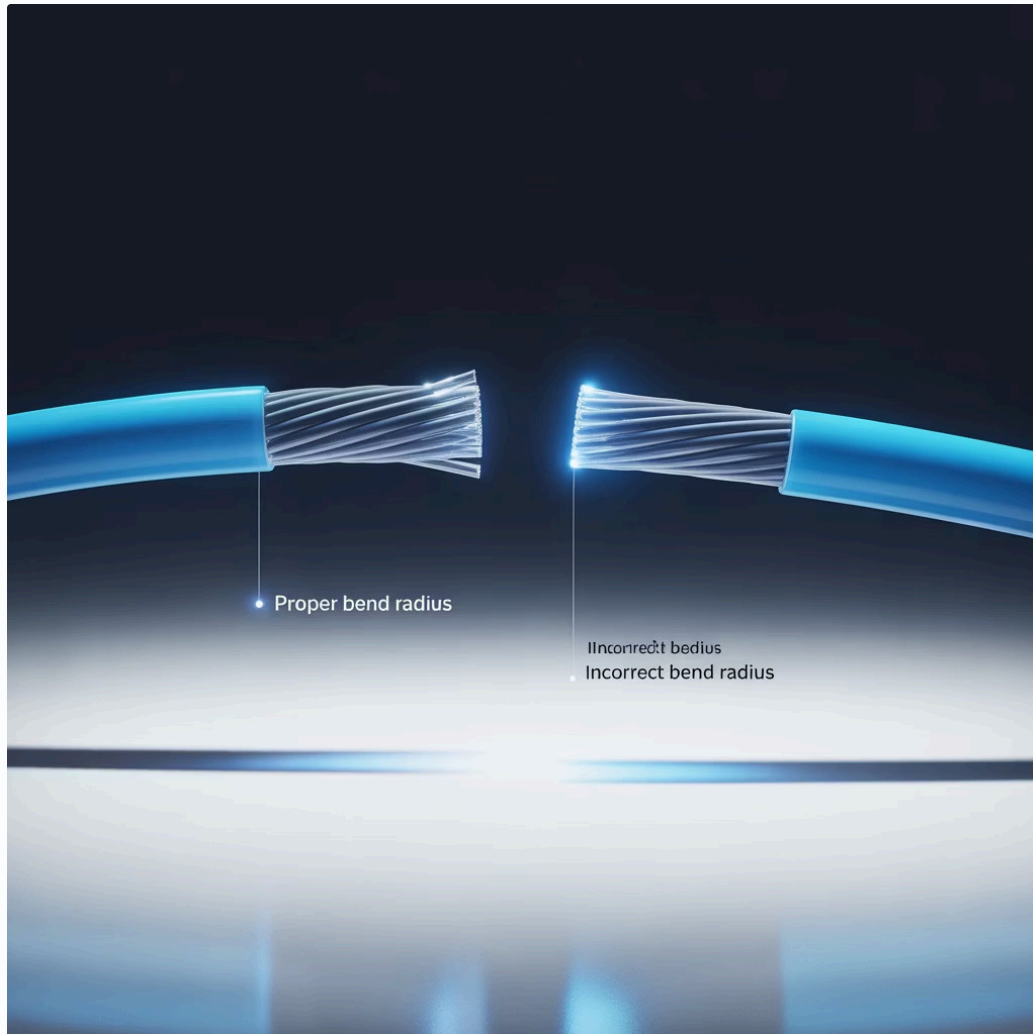
Example

3 CRC errors / 1000 frames = **0.3%**.

Media Issues

Concepts

- Fiber bend radius & loss.



Formula

Optical Loss:

$$10 \log_{10} \left(\frac{P_{in}}{P_{out}} \right)$$

Example

2 mW in, 1 mW out → **3 dB loss**.



Why This Matters

Predict performance & avoid bottlenecks

Using formulas to calculate network capacity helps identify potential issues before they impact users.

Minimize collisions & errors

Understanding collision domains and error rates helps design more efficient networks.

Reduce troubleshooting time

Knowing what metrics to check and how to interpret them speeds up problem resolution.

Justify upgrades with math-based evidence

Quantitative data provides compelling evidence when requesting network improvements.